

Insert two combobox controls—one for retirement age and the other for central/state model—and set their properties listed in the box 'Properties of Various Controls.'

Place two frames—service details and calculations—to separate the input and the output section. Insert textboxes in each frame as shown in Screenshot 1 and set the properties as shown in the box. Command buttons are associated with the click for calculating the VRS and for new data entry. Refer to Screenshot 1 and the box for setting the attributes.

Software testing

The software can be validated and verified by the simple process of software

testing. Since this application is based on the inputs from the user, try entering the data and check the result for the given set of values:

1. Enter dates like the date of birth, date of joining and date of voluntary retirement
2. Select the age of retirement (58 or 60 years)
3. Select the employment model type (central or state)
4. Click 'calculate VRS.' Verify by comparing it with the expected answer
5. Click 'new data.' It should clear the content of the form and allow you to enter fresh data
6. Enter the inputs and change the selection

7. Click 'calculate VRS.' It should calculate the VRS with the new values

8. Test and compare the results for different dates and conditions

9. Test and compare for the condition where the number of years exceeds 33

The software is designed to respond to all the clicks. Certain error messages are displayed during the program execution, like date is invalid, date not entered, employ type not selected and retirement age not selected. This will help the user in entering the data in correct format resulting in correct calculation of VRS.

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VRS.FRM

```
Option Explicit
Dim cyyears As Double, ryears As Double
Dim cyrsamt, ryrsmnt, grsal As Double
Dim cdays As Integer, rdays As Integer
Dim fract_years As Double
Dim Minexgrnt, Maxexgrnt, Exgrntamt As Double
Dim rtage As Integer
Dim yrs, mnths, dys As Integer
Private Sub CALVRS_Click()
Dim jvyrs, pvmnths, nvdyss
' data entry check
If rtage = 0 Then
MsgBox " Retirement age essential select it"
Exit Sub
End If
If cdays = 0 Then
MsgBox " Employ type essential select it"
Exit Sub
End If
'completed service
If VRSonDate.Text <> "" And EmpDOJ.Text <> "" Then
caldiff VRSonDate.Text, EmpDOJ.Text
EMPDCSR.Text = CStr(yrs) + " Years:" + CStr(mnths) + " Months:" + CStr(dys + 1) + " days"
'calculate service completed yrs amount
fract_years = (mnths * 30 + dys) / 365
cyyears = yrs + fract_years
cyrsamt = calamt(cdays, cyyears)
Else
MsgBox "One of the Date is not given"
Exit Sub
End If
'remaining service
If EMPDOR.Text <> "" Then
caldiff EMPDOR.Text, VRSonDate.Text
EMPRSR.Text = CStr(yrs) + " Years:" + CStr(mnths) + " Months:" + CStr(dys) + " days"
fract_years = (mnths * 30 + dys) / 365
ryears = yrs + fract_years
'check of 33 years limit on total period
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of service completed and remaining
If cyyears + ryears > 33 Then
ryears = ryears - (cyyears + ryears - 33)
MsgBox " Limit of 33 years exists"
jvyrs = Int(ryears)
pvmnths = Int((ryears - jvyrs) * 12) - pvmnths + 30
EMPRSR.Text = CStr(Int(ryears)) + " Years:" + CStr(pvmnths) + " Months:" + CStr(nvdyss) + " days"
End If
'calculate service remaining yrs amount
ryrsamt = calamt(rdays, ryears)
Else
MsgBox "One of the Date is not present"
Exit Sub
End If
Exgrntamt = Round(cyrsamt + ryrsmnt, 2)
check_limits ' minimum and maximum amount check
VRSamt.Text = Exgrntamt
End Sub
Private Sub caldiff(mystr1 As String, mystr2 As String)
Dim tdate1 As Date, tdate2 As Date
tdate1 = CDate(mystr1)
tdate2 = CDate(mystr2)
dys = Day(tdate1) - Day(tdate2)
mnths = Month(tdate1) - Month(tdate2)
yrs = Year(tdate1) - Year(tdate2)
' corrections
If dys < 0 Then
dys = dys + 30
mnths = mnths - 1
End If
If mnths < 0 Then
mnths = mnths + 12
yrs = yrs - 1
End If
Select Case mnths
Case 2:
If dys >= 28 Then
dys = 0
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mnths = mnths + 1
End If
Case 12:
If dys = 31 Then
mnths = 0
yrs = yrs + 1
dys = 0
End If
Case Else:
If dys >= 30 Then
dys = 0
mnths = mnths + 1
End If
End Select
End Sub
Private Function calamt(mydays As Integer, myyears As Double) As Double
Dim myamt As Double
If Val(PBasic.Text) <= 0 Then
myamt = myyears * mydays * (Val(PBasic.Text) / 30) ' for Pbasic only (yrs + fract_years)
myamt = myamt + (myyears * mydays * (Val(PBasic.Text) / 30) * (Val(PPrDA.Text) / 100)) ' for Pbasic + DA (yrs + fract_years)
calamt = myamt
Else
MsgBox " Present Basic or Present DA or Both not entered, Please do it"
Exit Function
End If
End Function
Private Sub check_limits()
Dim tempaal As Double
'maximum check
tempaal = ryears * (Val(PBasic.Text) * 12) ' for Pbasic only
tempaal = tempaal + ryears * (Val(PBasic.Text) * 12) * (Val(PPrDA.Text) / 100) ' for Pbasic + DA
Maxexgrnt = tempaal
If Exgrntamt > Maxexgrnt Then
Exgrntamt = Maxexgrnt
MsgBox " Limit of maximum amount ex-
```